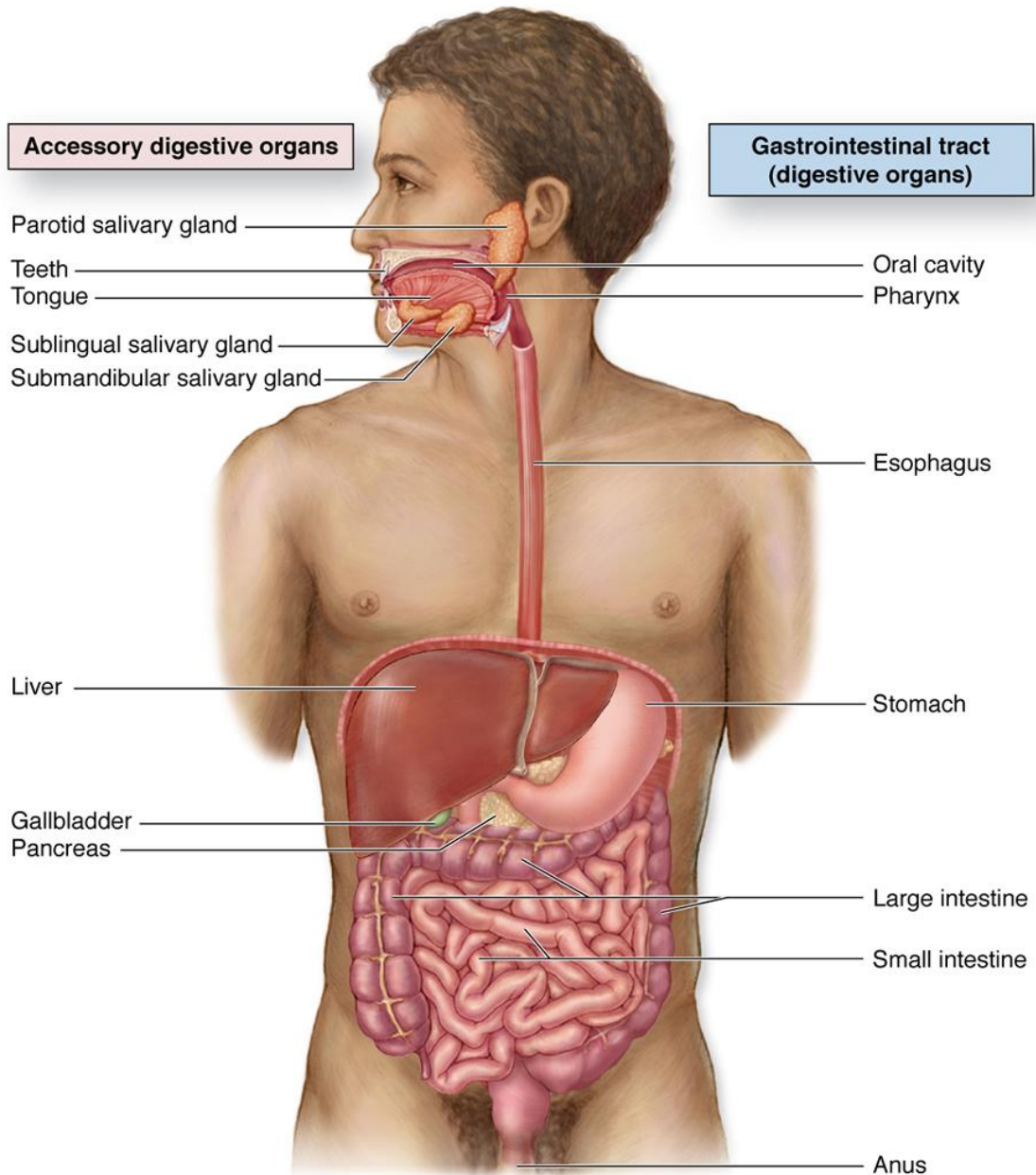


Digestive system



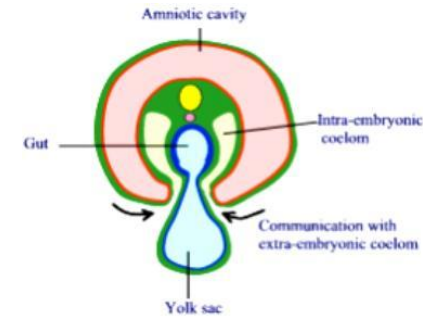
The **peritoneum** is the largest serous membrane of the human body.

Derives from lateral plates of mesoderm.

It has two layers:

1. parietal peritoneum
2. visceral peritoneum

They have a continuous mesothelial layer



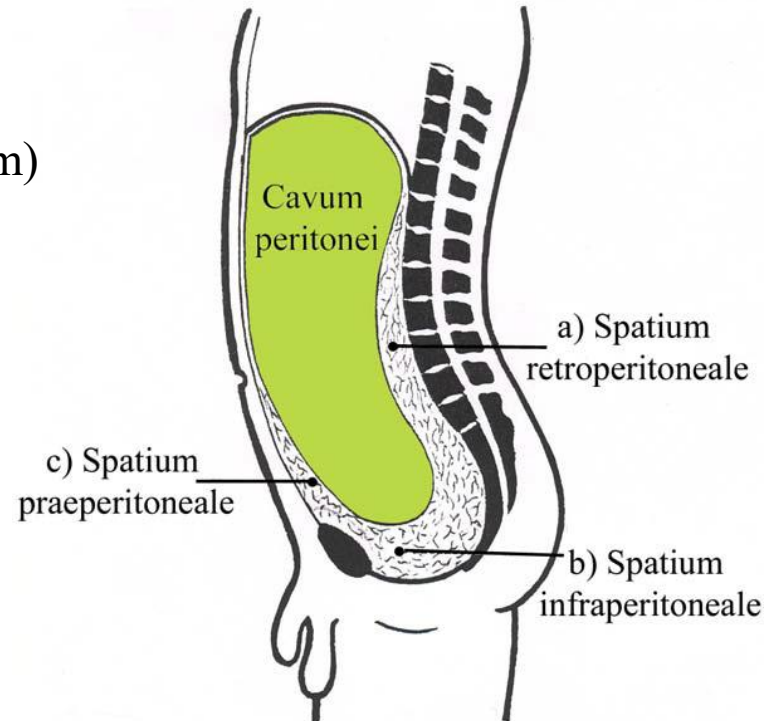
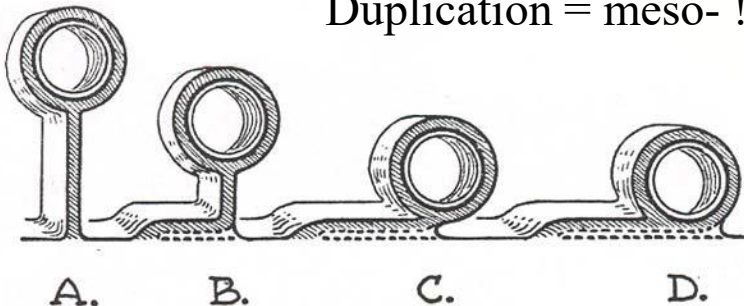
Abdominal and pelvic organs and structures –
according to their peritoneal relations - can be divided into 2 main groups:

Intraperitoneal - covered by peritoneum
and generally they have a duplication

Extraperitoneal (not surrounded by the peritoneum)
within this:

- Praeperitoneal
- Retroperitoneal (secunder and primary)
- Infracperitoneal structures.

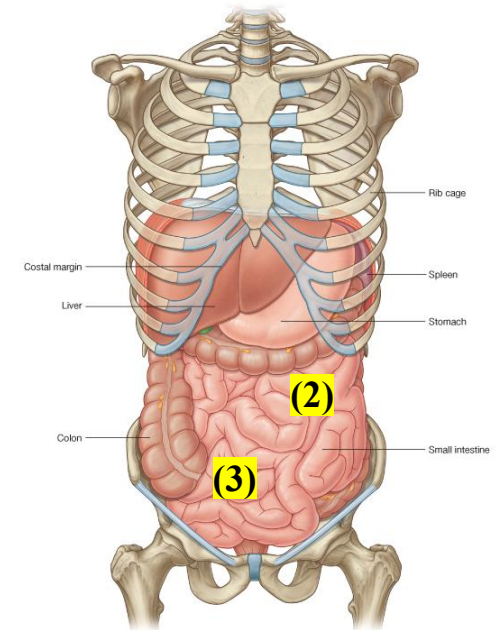
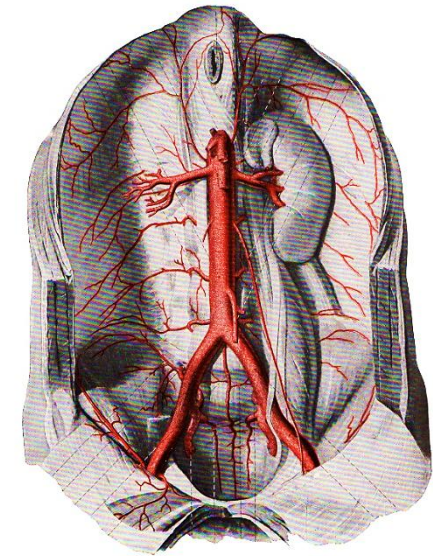
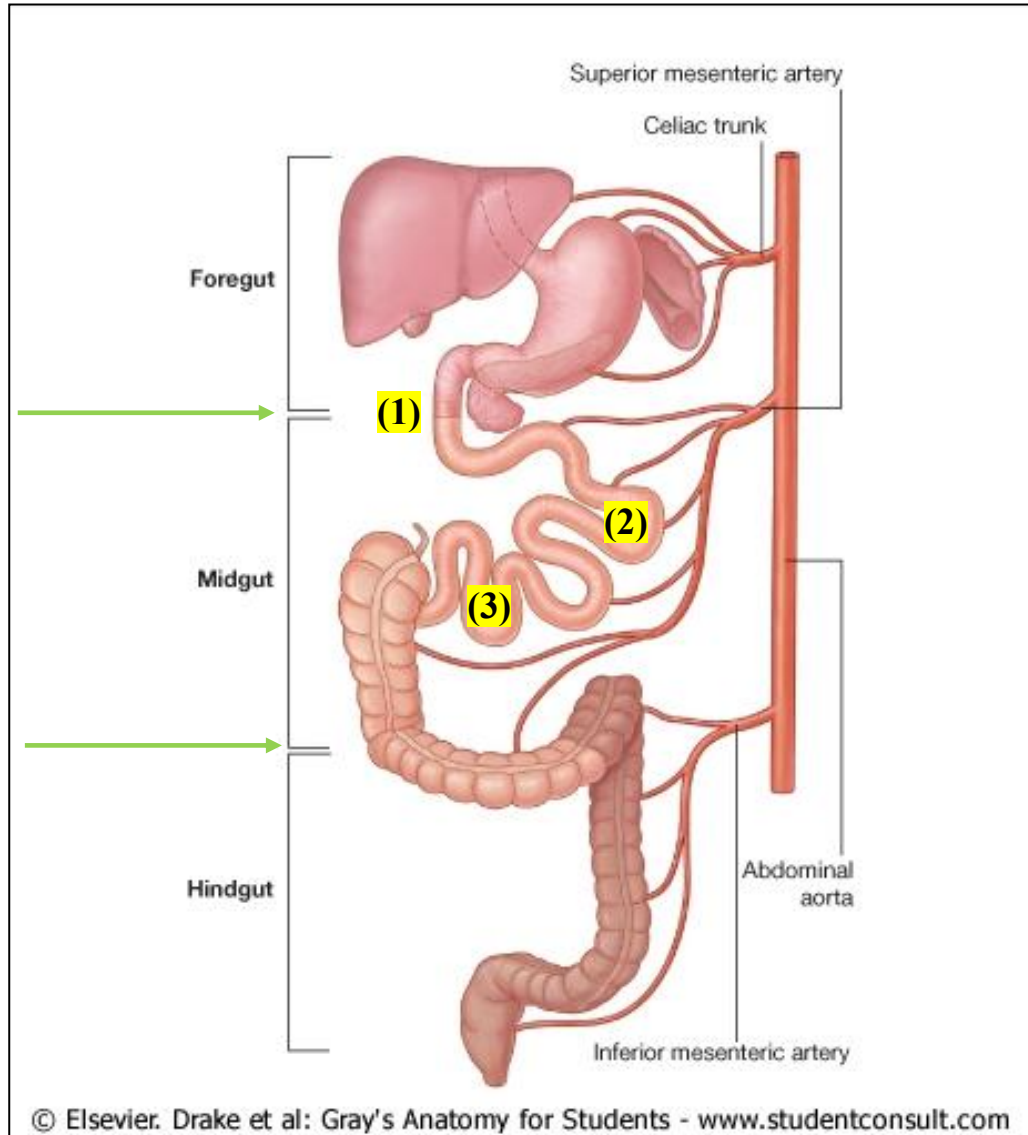
Duplication = meso- !!!!



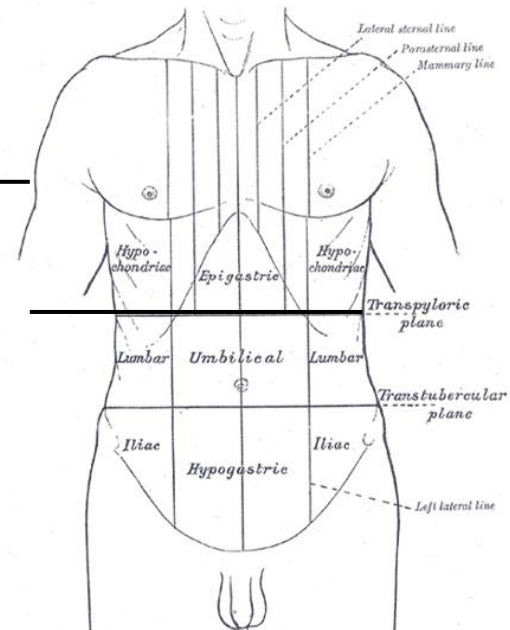
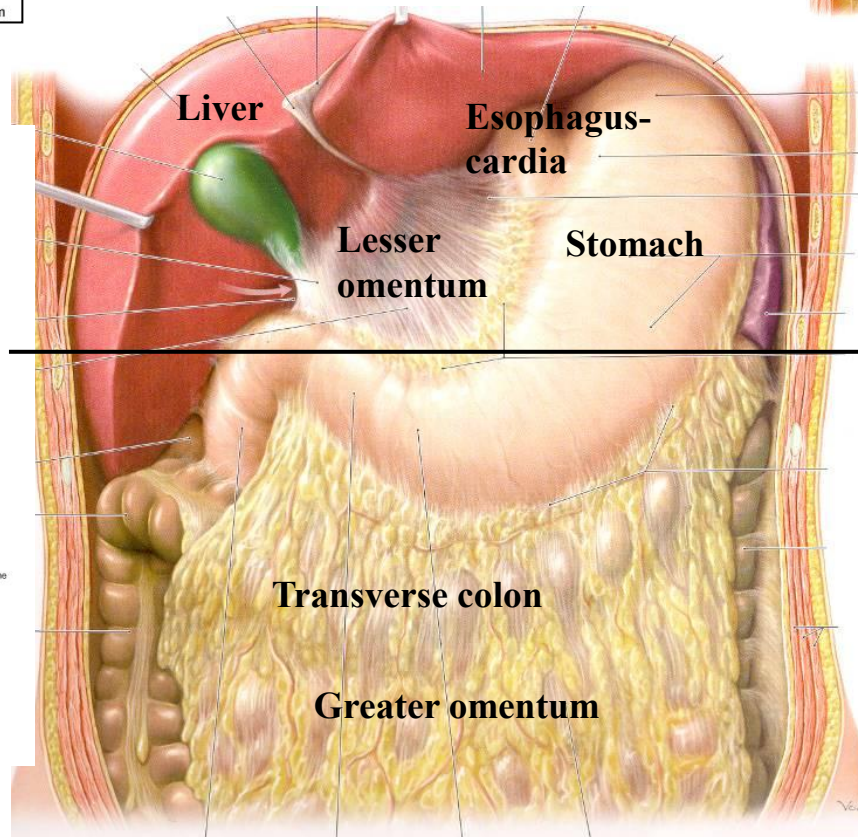
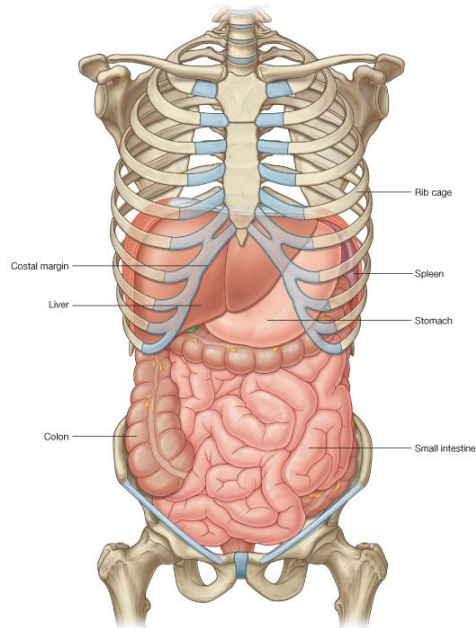
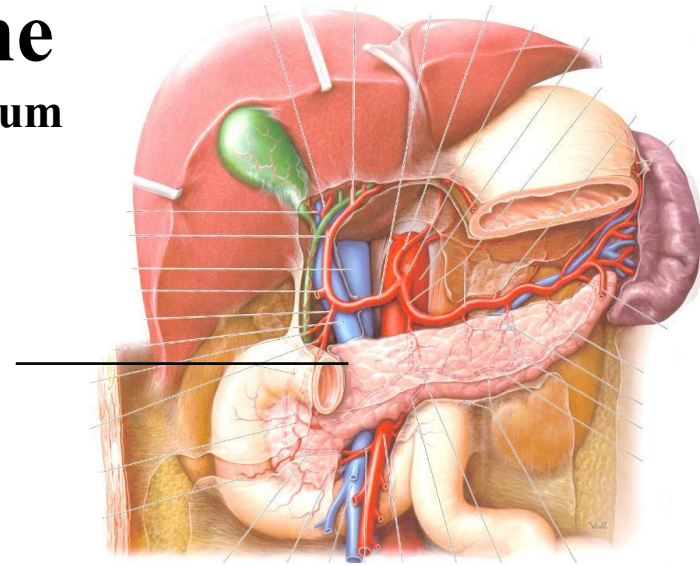
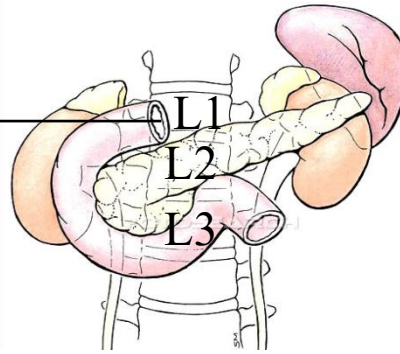
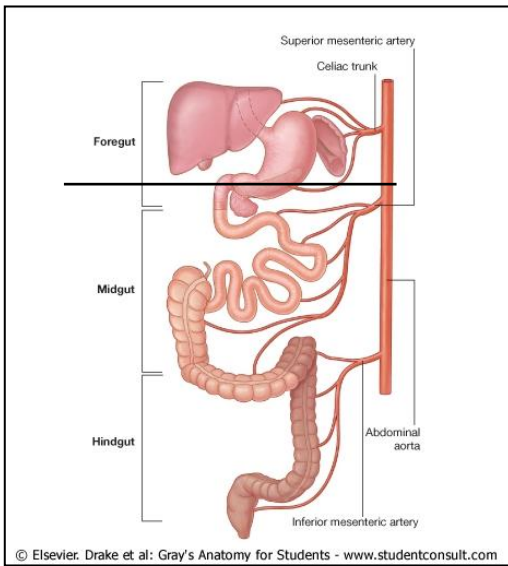
Abdominal part of the digestive system

developmental origin and blood supply (unpaired visceral branches of the abdominal aorta)

Small intestine: 1) duodenum, 2) jejunum, 3) ileum



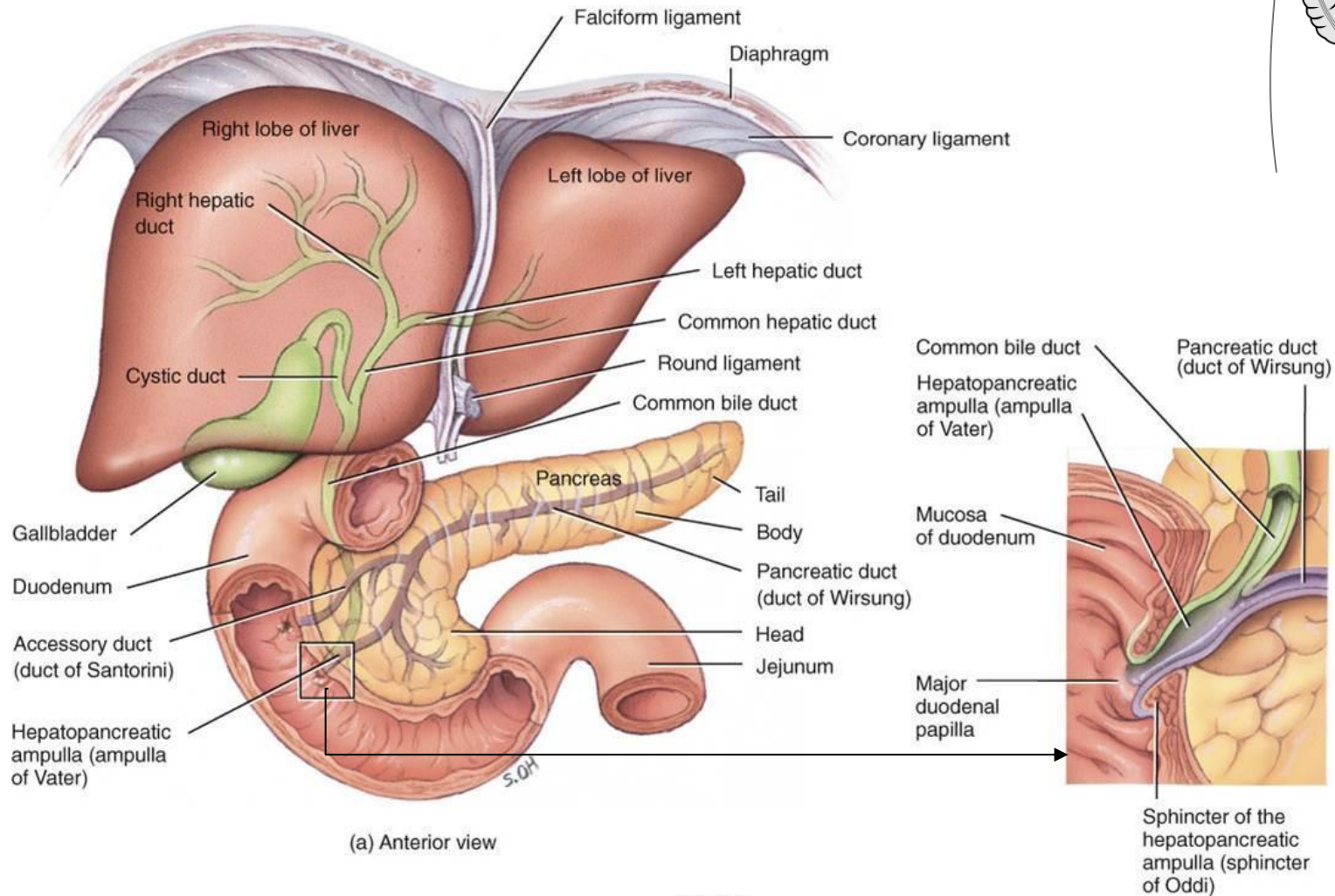
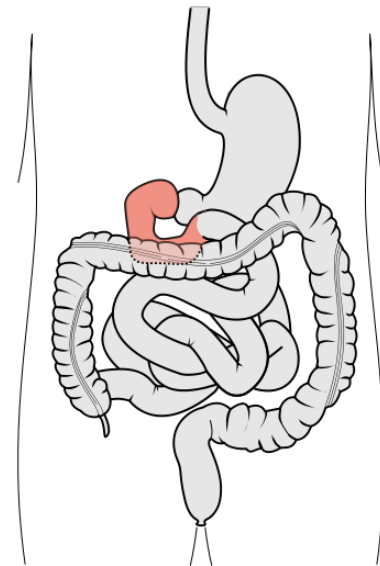
Transpyloric plane and the position of the duodenum



Duodenum

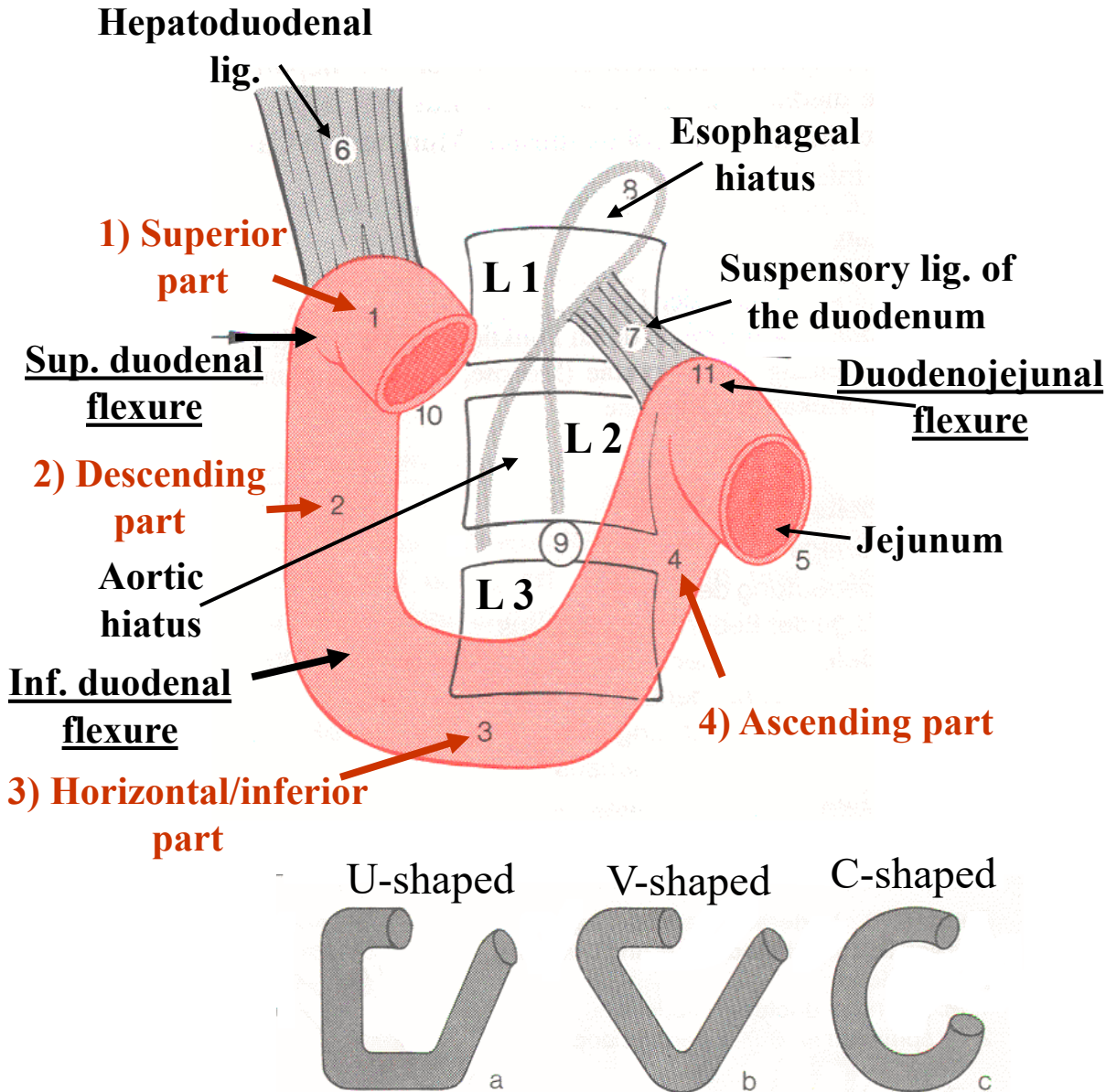
The duodenum is the proximal, U-shaped, and shortest (approx. 25–30 cm) segment of the small intestine, interconnecting the pylorus with the jejunum. *The name derived from the Latin *interstinum duodenum digitorum* ("12 fingers interstine").*

It acts as the primary site for mixing chyme with **bile** and **pancreatic enzymes** for nutrient digestion. Neutralizes stomach acid and initiates nutrient absorption.



(a) Anterior view

Parts of the duodenum and topography



Total length: 25-30 cm;

1) Superior part: initial segment=bulbus (or ampulla); moveable; intraperitoneal (hepatoduodenal lig.);

2) Descending part: fixed; retroperitoneal;

3) Horizontal/inferior part: fixed; retroperitoneal;

4) Ascending part: retro-, and intraperitoneal; The suspensory muscle and ligament suspends it to the origin of the sup. mesenteric artery.

Connections of the duodenum, the bile tract and the pancreas

The **major duodenal papilla (papilla of Vater)** is a small, nipple-like structure in the duodenum that acts as the primary entrance for **bile and pancreatic juice**, surrounded by the sphincter of Oddi, it regulates the flow of digestive enzymes into the duodenum.

The **minor duodenal papilla** serves as the entrance for the **accessory pancreatic duct**.

